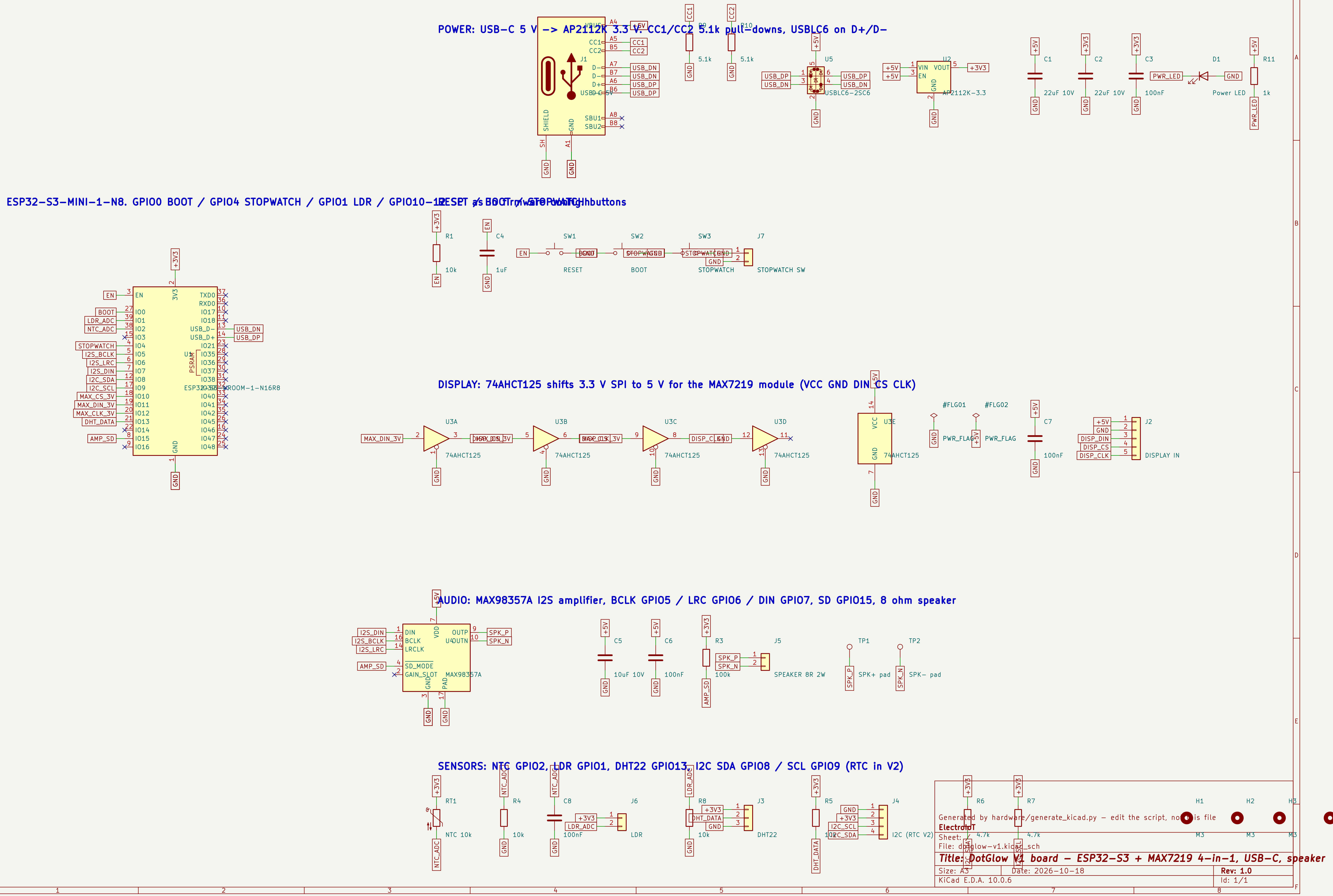
J1
VBUS
A4
+5V
CC1
A5
CC2
B5
D-
A7
USB_DN
D-
B7
D+
A6
USB_DP
B6
USB C 5V
SHIELD
SH
GND
A1
SBU1
A8
SBU2
B8
CC1
R9
5.1k
CC2
R10
5.1k
U5
USBLC6-2SC6
USB_DP
USB_DN
U2
VIN
VOUT
EN
AP2112K-3.3
+3V3
C1
22uF 10V
C2
22uF 10V
C3
100nF
D1
PWR_LED
Power LED
R11
1k
EN
BOOT
LDR_ADC
NTC_ADC
STOPWATCH
I2S_BCLK
I2S_LRC
I2S_DIN
I2C_SDA
I2C_SCL
MAX_CS_3V
MAX_DIN_3V
MAX_CLK_3V
DHT_DATA
AMP_SD
U1
PSRAM
ESP32-S3-WROOM-1-N16R8
3V3
TXD0
RXD0
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
R1
10k
C4
1uF
SW1
RESET
SW2
BOOT
SW3
STOPWATCH
J7
STOPWATCH SW
U3A
U3B
U3C
U3D
U3E
74AHCT125
MAX_DIN_3V
DISP_DIN
MAX_CS_3V
DISP_CS
MAX_CLK_3V
DISP_CLK
VCC
#FLG01
#FLG02
PWR_FLAG
C7
100nF
J2
DISPLAY IN
U4
DIN
BCLK
LRCLK
SD_MODE
GAIN_SLOT
VDD
OUTP
OUTN
PAD
MAX98357A
SPK_P
SPK_N
C5
10uF 10V
C6
100nF
R3
100k
J5
SPEAKER 8R 2W
TP1
SPK+ pad
TP2
SPK- pad
RT1
NTC 10k
R4
10k
C8
100nF
J6
LDR
R8
10k
J3
DHT22
R5
10k
J4
I2C (RTC V2)
R6
4.7k
R7
4.7k
H1
H2
H3
M3

POWER: USB-C 5 V -> AP2112K 3.3 V. CC1/CC2 5.1k pull-downs, USBLC6 on D+/D-

ESP32-S3-MINI-1-N8. GPIO0 BOOT / GPIO4 STOPWATCH / GPIO1 LDR / GPIO10-1

RESET / BOOT / STOPWATCH push buttons

DISPLAY: 74AHCT125 shifts 3.3 V SPI to 5 V for the MAX7219 module (VCC GND DIN CS CLK)

AUDIO: MAX98357A I2S amplifier, BCLK GPIO5 / LRC GPIO6 / DIN GPIO7, SD GPIO15, 8 ohm speaker

SENSORS: NTC GPIO2, LDR GPIO1, DHT22 GPIO13, I2C SDA GPIO8 / SCL GPIO9 (RTC in V2)

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ElectroIoT
Sheet: /
File: dotglow-v1.kicad_sch
Title: DotGlow V1 board – ESP32-S3 + MAX7219 4-in-1, USB-C, speaker
Size: A3 Date: 2026-10-18 **Rev: 1.0**
KiCad E.D.A. 10.0.6 Id: 1/1